

A Proof of the Dominated Bilinear Forms Tensor Conjecture

FA Banach run: `fa_banach_001`

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Source Problem

The source paper is:

G. Botelho, D. Pellegrino, and P. Rueda, *Dominated bilinear forms and 2-homogeneous polynomials*, arXiv:0905.2079.

The final conjecture of the paper asks whether, for an infinite-dimensional Banach space X ,

$$\mathcal{L}_{d,1}({}^2X) = \mathcal{L}({}^2X) \implies X \widehat{\otimes}_{\pi} X = X \widehat{\otimes}_{\varepsilon} X.$$

Here $\mathcal{L}_{d,1}({}^2X)$ denotes the continuous scalar-valued bilinear forms on $X \times X$ which are 1-dominated.

$$\begin{aligned}
&= \sup \left\{ \sum_{j=1}^n \|x_j\|_{w,r}^q, \sum_{j=1}^n \|x_j\|_{w,r}^q \right\} \\
&\leq \|P\|_{as(q;r)}^q (\|(x_j)_{j=1}^n\|_{w,r})^{mq} \\
&\leq K^q (\|(x_j)_{j=1}^n\|_{w,r})^{mq}.
\end{aligned}$$

It follows that

$$\left(\sum_{j=1}^n \|x_j\|_{w,r}^{\frac{s}{s-1}mq} \right)^{\frac{1}{(\frac{s}{s-1})mq}} \leq K^{\frac{1}{m}} \cdot \|(x_j)_{j=1}^n\|_{w,r}.$$

Since $\frac{s}{s-1}mq = \frac{mq}{1-q}$, n and $x_1, \dots, x_n \in X$ are arbitrary, we conclude that id_X is $(\frac{mq}{1-q}, r)$ -summing. $\square$

The following theorem holds for spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$:

Theorem 2.7. *If $\cot X = q > 2$, then $\mathcal{P}_{d,r}(^2X) \neq \mathcal{P}(^2X)$ for $1 \leq r < q^*$, where q^* is the conjugate of q . In other words, $\mathcal{P}_X \geq q^*$.*

Proof. Real case: Let $1 \leq r < q^*$. Combining Lemma 2.1 and [9, Corollary 11.16(b)] it is immediate that $\mathcal{P}_{d,r}(^2X) = \mathcal{P}_{d,1}(^2X)$. If $\mathcal{P}_{d,r}(^2X) = \mathcal{P}_{d,1}(^2X) = \mathcal{P}(^2X)$, from Theorem 2.6 we could conclude that id_X is $(2; 1)$ -summing, but this is impossible because $\cot X > 2$.

Complex case: If X is a complex Banach space, $\cot X = q > 2$ and $1 \leq r < q^*$, by [3, Lemma 3.1] we know that $\cot X_{\mathbb{R}} = q > 2$, so there is a non- r -dominated polynomial $P \in \mathcal{P}(^2X_{\mathbb{R}})$. Denoting by $\tilde{P}$ the complexification of P we have that $\tilde{P} \in \mathcal{P}(^2X)$ and following the lines of [16, Proposition 4.30] it is not difficult to prove that $\tilde{P}$ fails to be r -dominated either. $\square$

Remark 2.8. Let X be any of the spaces constructed by Pisier [18]. By Example 2.3 we know that $\mathcal{P}_{d,1}(^2X) = \mathcal{P}(^2X)$, which makes clear that Theorem 2.7 is sharp in the sense that it is not valid for cotype 2 spaces.

Conjecture. We conjecture that if X is infinite-dimensional and $\mathcal{L}_{d,1}(^2X) = \mathcal{L}(^2X)$, then $X \tilde{\otimes}_{\pi} X = X \tilde{\otimes}_{\varepsilon} X$. Observe that for an infinite-dimensional space X with $\mathcal{L}_{d,1}(^2X) = \mathcal{L}(^2X)$ and $X \tilde{\otimes}_{\pi} X \neq X \tilde{\otimes}_{\varepsilon} X$, if any, we should have:

- X has no unconditional basis [4, Theorem 3.2];
- X has cotype $2 + \varepsilon$ for every $\varepsilon > 0$ (Proposition 2.4);

- X does not have cotype 2 (Proposition 2.2);
- X' is a GT space [11, Theorem 3.4];
- Every linear operator from X to X' is absolutely 1-summing (by [1, Lemma 3.4] this is a consequence of $\mathcal{L}_{d,1}(^2X) = \mathcal{L}(^2X)$), in particular X is Arens-regular;
- Not every linear operator from X to X' is integral (this is a consequence of $X \tilde{\otimes}_{\pi} X \neq X \tilde{\otimes}_{\varepsilon} X$).

References

Result

Theorem 1. *Let X be a Banach space. If every continuous scalar-valued bilinear form on $X \times X$ is 1-dominated, then every continuous scalar-valued bilinear form on $X \times X$ is integral. Consequently*

$$X \widehat{\otimes}_{\pi} X = X \widehat{\otimes}_{\varepsilon} X.$$

In particular, the conjecture from arXiv:0905.2079 is true. The infinite-dimensional hypothesis is not needed.

Proof Intuition

The obstruction listed in the source paper says that a counterexample would have an operator $T : X \rightarrow X^*$ which is not integral. We show that this cannot happen.

The hypothesis first implies that every operator $X \rightarrow X^*$ is absolutely 1-summing. This is already noted in the source paper, but the proof is short and is included below. The missing point is to make the adjoint

$$T^* : X^{**} \rightarrow X^*$$

absolutely 1-summing as well, because the standard duality theorem for integral operators says that T is integral exactly when T^* is 1-summing.

To prove this adjoint estimate, one uses local reflexivity. A finite family in X^{**} can be pulled back almost isometrically to X while preserving its values on finitely many functionals of the form Tx . The orientation matters: the estimate uses the transposed operator $T^{\tau}(y)(x) = T(x)(y)$, which is also absolutely 1-summing because the hypothesis applies to the transposed bilinear form.

Definitions and Standard Facts

For a finite sequence $(x_j)_{j=1}^n$ in X , write

$$\|(x_j)_{j=1}^n\|_{w,1} = \sup_{x^* \in B_{X^*}} \sum_{j=1}^n |x^*(x_j)|.$$

A bilinear form $A : X \times X \rightarrow \mathbb{K}$ is 1-dominated if there is a constant D such that, for all finite sequences (x_j) and (y_j) in X ,

$$\left(\sum_{j=1}^n |A(x_j, y_j)|^{1/2} \right)^2 \leq D \|(x_j)_{j=1}^n\|_{w,1} \|(y_j)_{j=1}^n\|_{w,1}. \quad (1)$$

We use two standard facts. First, the principle of local reflexivity: if $E \subset X^{**}$ and $F \subset X^*$ are finite-dimensional and $\delta > 0$, then there is a linear map $R : E \rightarrow X$ with $\|R\| \leq 1 + \delta$ such that

$$f(Rx^{**}) = x^{**}(f) \quad (x^{**} \in E, f \in F).$$

Second, the duality theorem for integral operators:

$$T : X \rightarrow Y \text{ is integral} \iff T^* : Y^* \rightarrow X^* \text{ is absolutely 1-summing.} \quad (2)$$

We cite these in the standard forms found in the literature on tensor norms and absolutely summing operators.

From Dominated Bilinear Forms to Summing Operators

Lemma 1. *Let $T : X \rightarrow X^*$ be bounded and set $A_T(x, y) = T(x)(y)$. If A_T is 1-dominated, then T is absolutely 1-summing.*

Proof. Let D be a constant for A_T in (1). Fix $x_1, \dots, x_n \in X$ and $\eta > 0$. Choose $y_j \in B_X$ so that

$$|T(x_j)(y_j)| \geq (1 - \eta)\|Tx_j\|.$$

Put $a_j = |T(x_j)(y_j)|$. If $\sum_j a_j = 0$, there is nothing to prove. Otherwise apply (1) to the two sequences (x_j) and $(a_j y_j)$. Since

$$\|(a_j y_j)_{j=1}^n\|_{w,1} \leq \sum_{j=1}^n a_j,$$

we get

$$\left(\sum_{j=1}^n |a_j T(x_j)(y_j)|^{1/2} \right)^2 \leq D \|(x_j)_{j=1}^n\|_{w,1} \sum_{j=1}^n a_j.$$

The left side is $(\sum_j a_j)^2$. Cancelling $\sum_j a_j$ gives

$$\sum_{j=1}^n a_j \leq D \|(x_j)_{j=1}^n\|_{w,1}.$$

Letting $\eta \downarrow 0$, we obtain

$$\sum_{j=1}^n \|Tx_j\| \leq D \|(x_j)_{j=1}^n\|_{w,1}.$$

Thus T is absolutely 1-summing. □

The Local Reflexivity Step

Lemma 2. *Assume every continuous bilinear form on $X \times X$ is 1-dominated. Then, for every bounded operator $T : X \rightarrow X^*$, the adjoint*

$$T^* : X^{**} \rightarrow X^*$$

is absolutely 1-summing.

Proof. Let $T : X \rightarrow X^*$ be bounded. Define the transposed operator

$$T^\tau : X \rightarrow X^*, \quad T^\tau(y)(x) = T(x)(y).$$

The bilinear form associated with T^τ is the transpose of A_T , so it is 1-dominated by hypothesis. By the preceding lemma, T^τ is absolutely 1-summing.

Let $x_1^{**}, \dots, x_n^{**} \in X^{**}$ and let $\eta > 0$. Choose $z_j \in B_X$ such that

$$\|T^* x_j^{**}\| \leq |(T^* x_j^{**})(z_j)| + \eta/n = |x_j^{**}(Tz_j)| + \eta/n.$$

Set

$$E = \text{span}\{x_1^{**}, \dots, x_n^{**}\} \subset X^{**}, \quad F = \text{span}\{Tz_1, \dots, Tz_n\} \subset X^*.$$

By local reflexivity, for every $\delta > 0$ there is a linear map $R : E \rightarrow X$ with $\|R\| \leq 1 + \delta$ and

$$f(Rx^{**}) = x^{**}(f) \quad (x^{**} \in E, f \in F).$$

In particular,

$$x_j^{**}(Tz_j) = (Tz_j)(Rx_j^{**}) = T^\tau(Rx_j^{**})(z_j).$$

Therefore

$$\begin{aligned} \sum_{j=1}^n \|T^* x_j^{**}\| &\leq \eta + \sum_{j=1}^n |T^\tau(Rx_j^{**})(z_j)| \\ &\leq \eta + \sum_{j=1}^n \|T^\tau(Rx_j^{**})\| \\ &\leq \eta + \pi_1(T^\tau) \sup_{x^* \in B_{X^*}} \sum_{j=1}^n |x^*(Rx_j^{**})| \\ &\leq \eta + \pi_1(T^\tau)(1 + \delta) \sup_{\psi \in B_{E^*}} \sum_{j=1}^n |\psi(x_j^{**})| \\ &= \eta + \pi_1(T^\tau)(1 + \delta) \|(x_j^{**})_{j=1}^n\|_{w,1}. \end{aligned}$$

Letting $\eta \downarrow 0$ and $\delta \downarrow 0$ gives

$$\sum_{j=1}^n \|T^* x_j^{**}\| \leq \pi_1(T^\tau) \|(x_j^{**})_{j=1}^n\|_{w,1}.$$

Thus T^* is absolutely 1-summing. □

Proof of the Theorem

Let $A : X \times X \rightarrow \mathbb{K}$ be a continuous bilinear form, and let

$$T_A : X \rightarrow X^*, \quad T_A(x)(y) = A(x, y),$$

be its associated operator. By the local reflexivity lemma,

$$T_A^* : X^{**} \rightarrow X^*$$

is absolutely 1-summing. By the integral-operator duality (2), T_A is integral. Equivalently, A is an integral bilinear form.

Hence every continuous scalar-valued bilinear form on $X \times X$ is integral. By the standard tensor-norm characterization of integral bilinear forms, this is equivalent to the identity map on the algebraic tensor product extending to an isomorphism

$$X \widehat{\otimes}_\pi X \cong X \widehat{\otimes}_\varepsilon X$$

given by the canonical tensor identification. In the notation of the source paper,

$$X \widehat{\otimes}_\pi X = X \widehat{\otimes}_\varepsilon X.$$

This proves the conjecture.

Verification Notes

The proof is non-computational. The only delicate point is the orientation in the local-reflexivity argument: $x^{**}(Tz)$ is represented as $(Tz)(Rx^{**})$, which is controlled by the transposed operator T^τ , not by T . Since the hypothesis applies to all bilinear forms, T^τ is absolutely 1-summing by the first lemma.

The argument does not use infinite dimensionality, cotype, unconditionality, or any approximation property.

Novelty Check

On 2026-06-28, I checked the lightweight run indexes for arXiv:0905.2079, “dominated bilinear”, “2-homogeneous”, and the tensor-product keywords. No prior packet or attempt for this conjecture was found. I also searched the local parsed arXiv source pool for the exact title and close dominated-bilinear phrases; this found only the source paper and the deterministic scan record.

Web searches on the same date included exact and close variants such as “all bilinear forms 1-dominated Banach space”, “Dominated bilinear forms and 2-homogeneous polynomials conjecture”, “ $T:X$ to X^* is 1-summing then T is integral”, and “absolutely 1-summing X^* local reflexivity”. No later explicit solution of the conjecture was found in that bounded search.

Human Review Recommendation

This should be reviewed as a candidate full solution. The main check is the standard operator-ideal step T integral iff T^* is absolutely 1-summing, and the local-reflexivity estimate showing that T^* is 1-summing using T^τ .

References

References

- [1] G. Botelho, D. Pellegrino, and P. Rueda, *Dominated bilinear forms and 2-homogeneous polynomials*, arXiv:0905.2079.
- [2] A. Defant and K. Floret, *Tensor Norms and Operator Ideals*, North-Holland Mathematics Studies 176, North-Holland, 1993.
- [3] J. Diestel, H. Jarchow, and A. Tonge, *Absolutely Summing Operators*, Cambridge University Press, 1995.